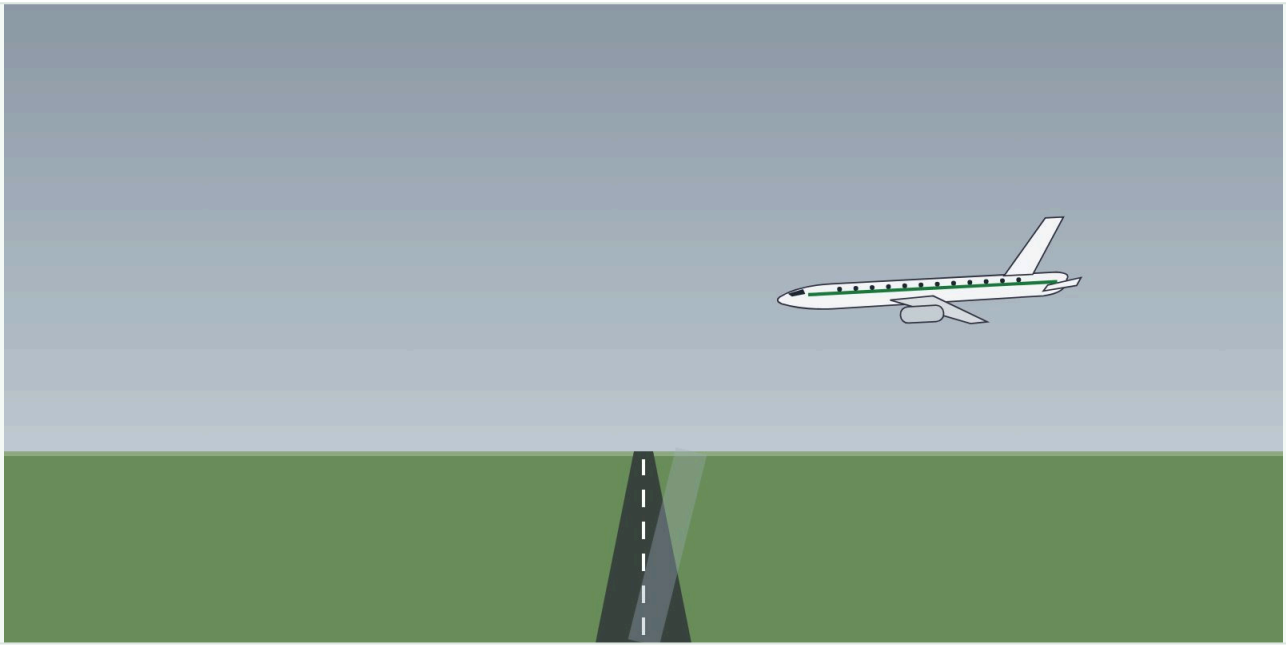
MODULE 1 · WIND AND LIMITATIONS

Lesson 02: Aircraft Wind Limitations

Lesson 2 of 3 · about 6 min

LEARNING OBJECTIVES

- Distinguish demonstrated crosswind from a limitation.
- Apply tailwind limits.
- Use the manufacturer's crosswind guidance by runway condition.



Crosswind limits decrease as the runway condition worsens.

Demonstrated crosswind

The AFM states the **maximum demonstrated crosswind** — the highest crosswind tested during certification. Unless the AFM says it is a limitation, it is **not** a structural or legal limit — but the company normally turns it, or a lower value, into an operational limit in the OM B.

Tailwind

The AFM gives a **maximum tailwind** for take-off and landing (commonly 10 kt; some aircraft are approved for more with specific conditions). This **is** a limitation. Performance must also be calculated with the reported tailwind.

Crosswind and runway condition

- Manufacturers publish recommended maximum crosswinds that **decrease as the runway condition worsens** — by runway condition code (RWYCC) or braking action.
- On a contaminated runway, use the crosswind for the reported RWYCC (see the Cold Weather Operations course).

Other wind-related limits

- **Autoland and low-visibility operations** have their own headwind, tailwind and crosswind limits.
- Doors: maximum wind for opening or closing passenger and cargo doors.
- Engine start in strong tailwind or crosswind (especially for some engines and propellers).

COMPANY LIMITS

The actual Domestic Airlines wind limits for each fleet — crosswind by runway condition, tailwind, gusts, limits for the first officer as pilot flying and for low-visibility / autoland — are in the **OM B**, the **FCOM / AOM** and the **AFM**. Always use them; the figures in this course are examples only.

KEY TAKEAWAYS

- Demonstrated crosswind: tested, not normally a limitation — but the company may make it one.
- Tailwind limit in the AFM is a limitation.
- Crosswind limits reduce with the runway condition; autoland has its own limits.

KNOWLEDGE CHECK

The AFM maximum demonstrated crosswind is:

A. Always a structural limit

B. The highest crosswind tested in certification — not a limitation unless the AFM says so

C. The same as the tailwind limit

D. Only for take-off

Answer: B. It is a demonstrated value, not a limitation unless stated.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.